

The reactive oxygen species-total antioxidant capacity score is a new measure of oxidative stress to predict male infertility.

The imbalance between reactive oxygen species (ROS) production and total antioxidant capacity (TAC) in seminal fluid indicates oxidative stress and is correlated with male infertility. A composite ROS-TAC score may be more strongly correlated with infertility than ROS or TAC alone. We measured ROS, TAC, and ROS-TAC scores in semen from 127 patients and 24 healthy controls. Of the patients, 56 had varicocele, eight had varicocele with prostatitis, 35 had vasectomy reversals, and 28 had idiopathic infertility. ROS levels were higher among infertile men, especially those with varicocele with prostatitis (mean $\pm$ SE, 3.25 $\pm$ 0.89) and vasectomy reversals (2.65 $\pm$ 1.01). All infertile groups had significantly lower ROS-TAC scores than control. ROS-TAC score identified 80% of patients and was significantly better than ROS at identifying varicocele and idiopathic infertility. The 13 patients whose partners later achieved pregnancies had a mean ROS-TAC score of 47.7 $\pm$ 13.2, similar to controls but significantly higher than the 39 patients who remained infertile (35.8 $\pm$ 15.0; $P < 0.01$). ROS-TAC score is a novel measure of oxidative stress and is superior to ROS or TAC alone in discriminating between fertile and infertile men. Infertile men with male factor or idiopathic diagnoses had significantly lower ROS-TAC scores than controls, and men with male factor diagnoses that eventually were able to initiate a successful pregnancy had significantly higher ROS-TAC scores than those who failed.